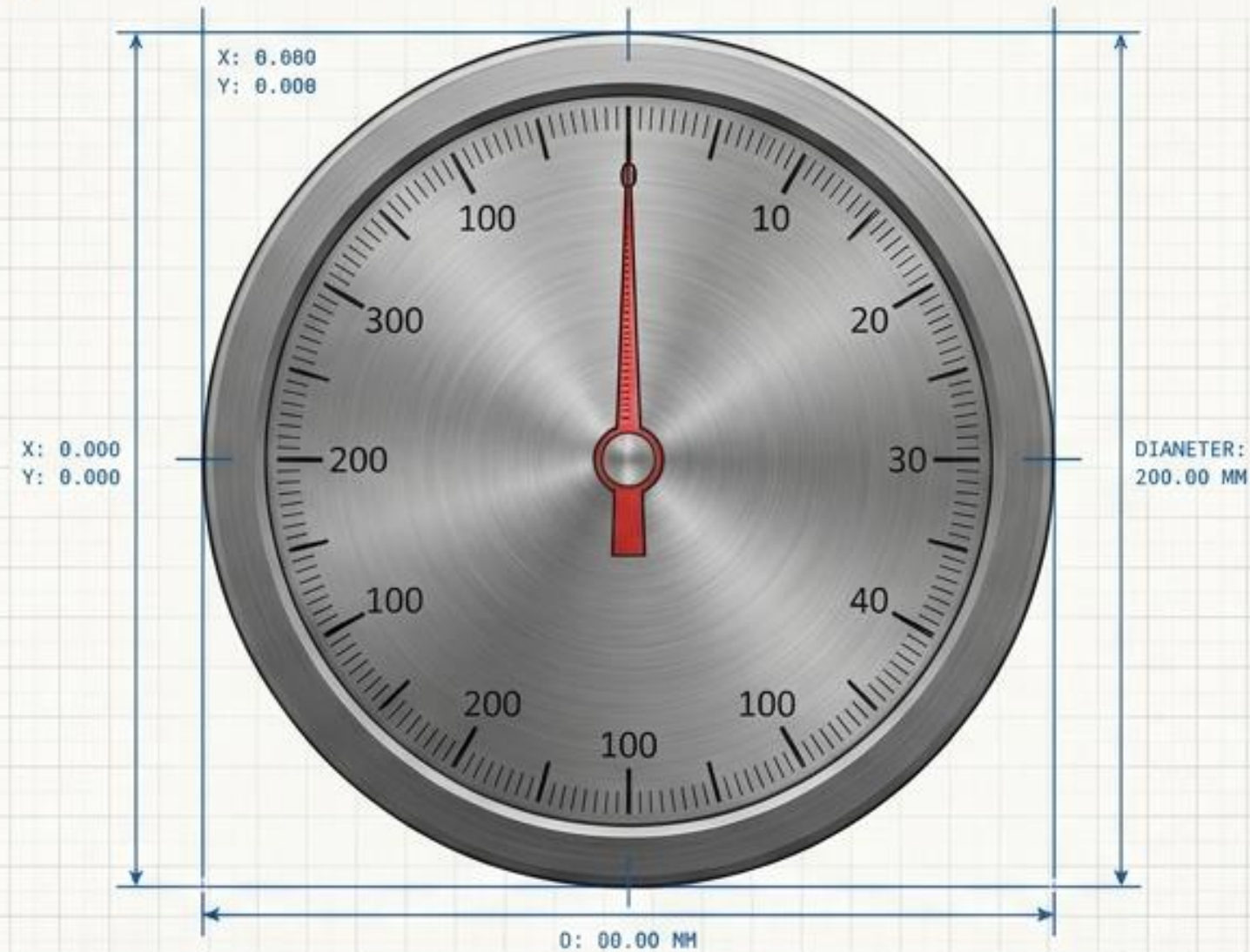


Operating Parameters of AI Environments

A Visual Blueprint to the 6 Dimensions of Task Complexity

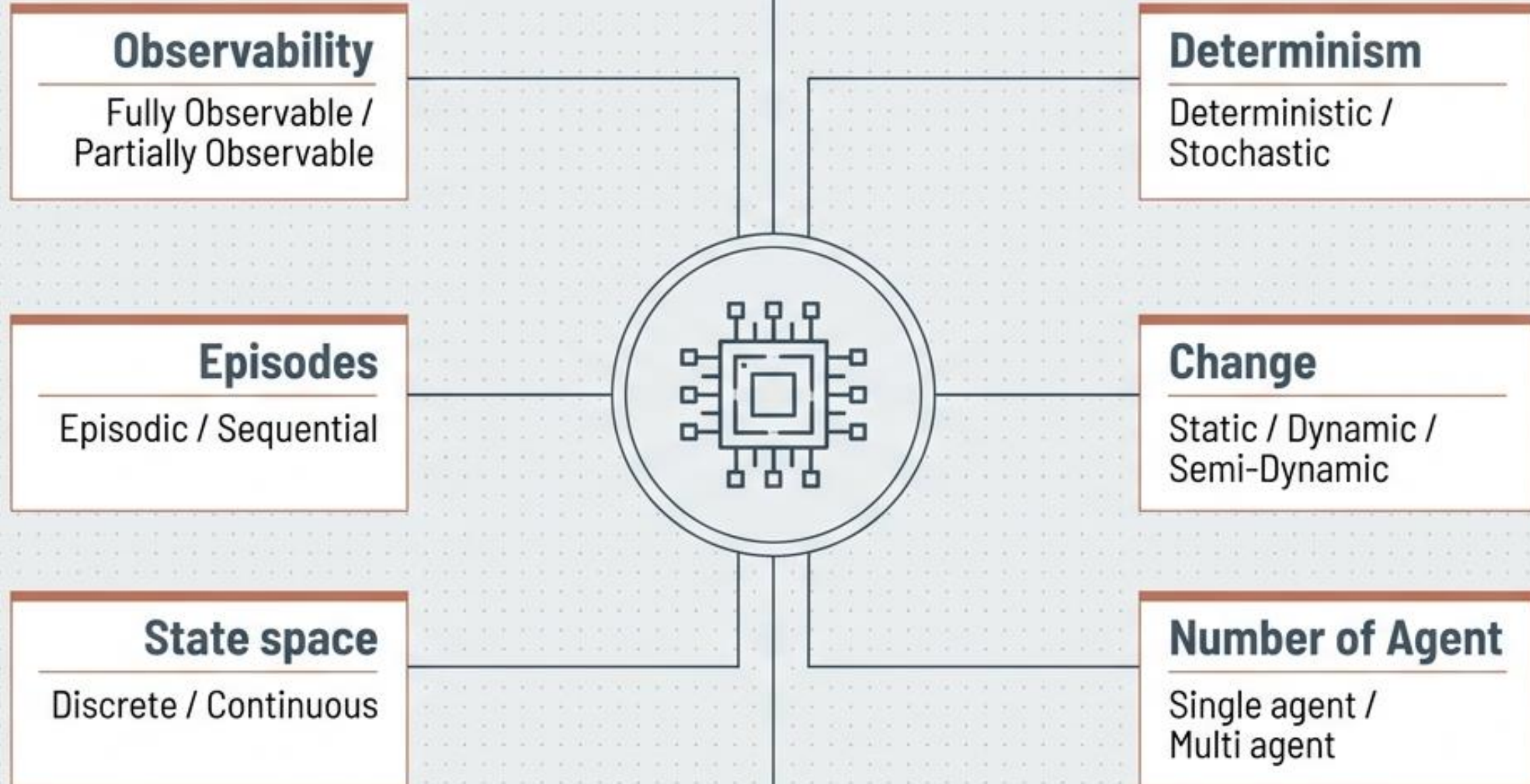


Instructor: Engr. Arifa

Department of Artificial Intelligence

Learning Outcomes

Distinguish different task-environment properties such as observable/partially observable, deterministic/stochastic, and static/dynamic.



Observability dictates what the agent's sensors can detect

Fully Observable Environment



An environment is fully observable if the agent's sensors provide complete and accurate information about the current state.

Characteristics:

- Complete information available
- No hidden state
- Easier decision making

Example: Chess (All pieces are visible)

Partially Observable Environment



An environment is partially observable if the agent cannot observe the complete state of the environment.

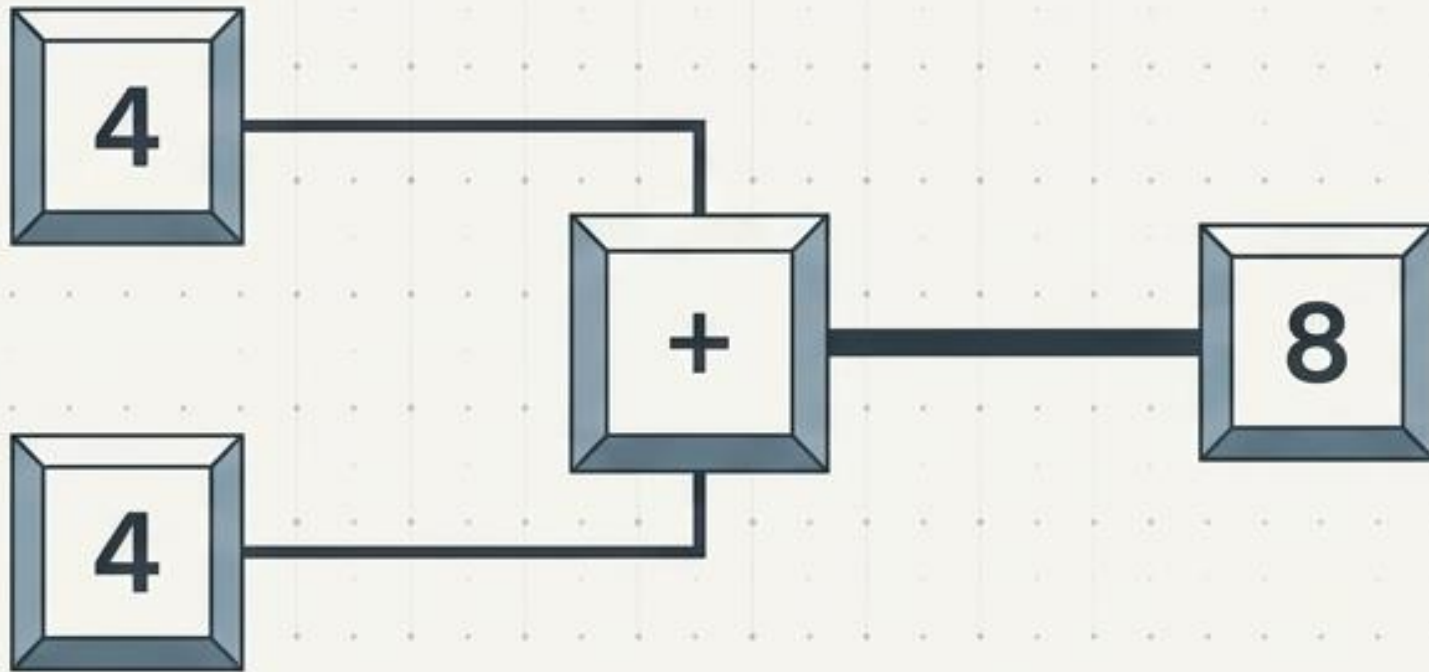
Characteristics:

- Incomplete information
- Hidden state
- Requires memory or inference

Example: Driving a car in heavy fog (The agent cannot see all the objects on the road)

Determinism measures the predictability of outcomes

Deterministic

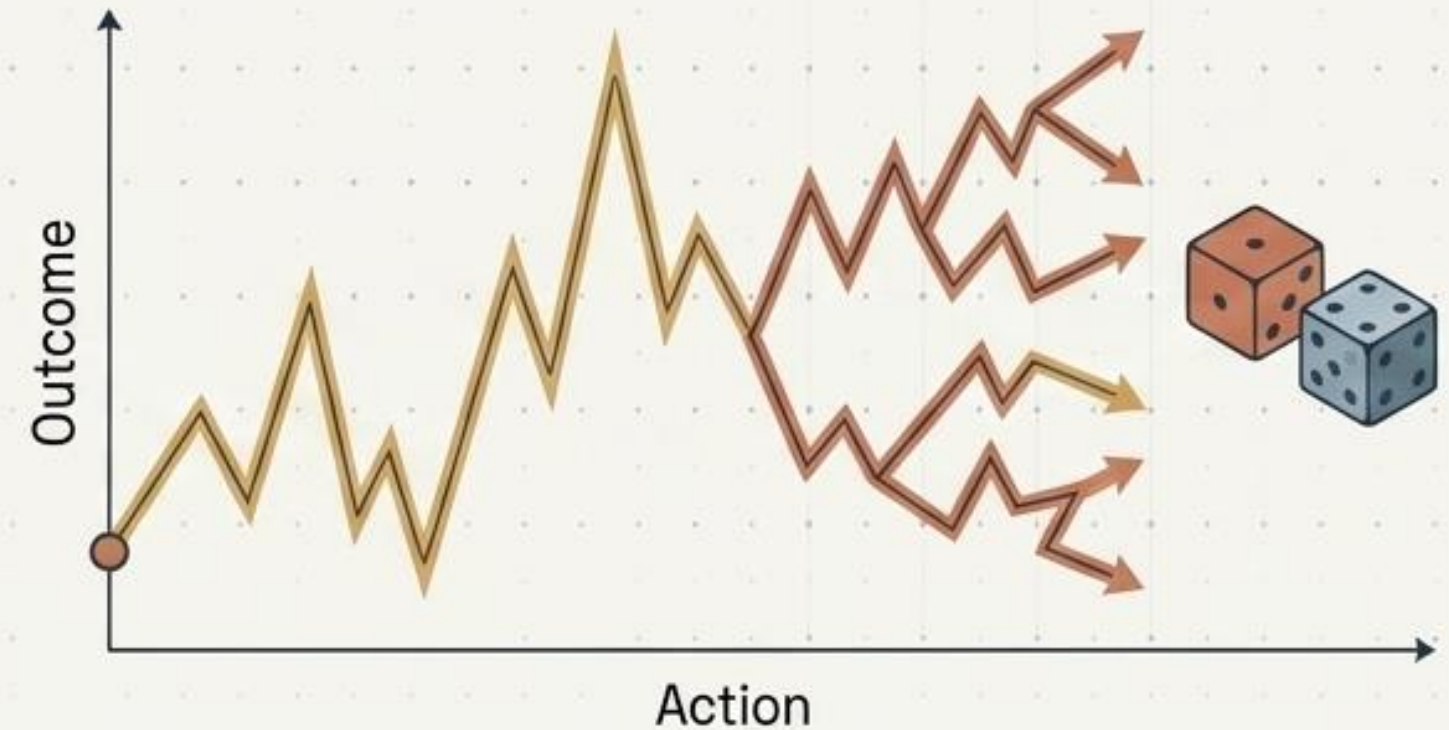


If an action always produces the same predictable result, the environment is deterministic.

Rules: Input is fixed | Output is fixed |

Example: Solving the mathematical equation $4 + 4 = 8$

Stochastic (Non-Deterministic)



If the action or result is uncertain or influenced by randomness, the environment is stochastic.

Rules: The same action produces different outcomes |

Example: Stock market prediction | Dice

Knowledge Check 1: The origins of environmental uncertainty

$$\left[\text{Partially Observable} \right] \text{ AND/OR } \left[\text{Stochastic} \right] = \boxed{\text{UNCERTAINTY}}$$

Core Rule:

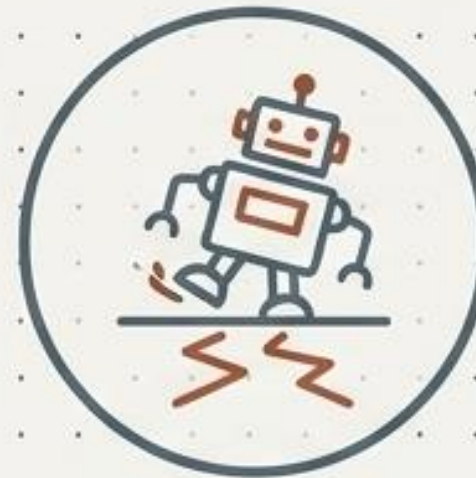
An uncertain environment is one in which the agent cannot predict the exact state of the environment or the outcomes of its action.

Conditions:

Uncertainty arises if either of the following conditions exist:
Partially observable and non-deterministic (stochastic).
Therefore, an environment becomes uncertain if it is either not fully observable or non-deterministic.

Real-World Examples:

Driving in dense fog where the driver cannot clearly see surrounding vehicles.

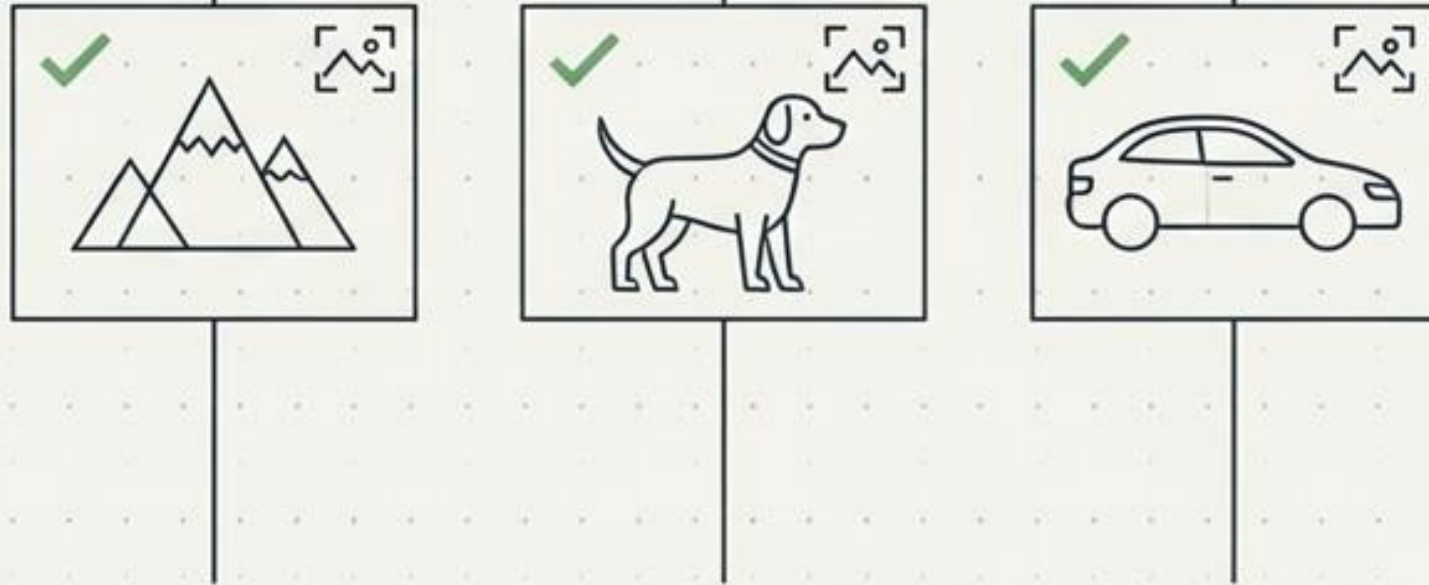


Real-World Examples:

A robot moving on a slippery floor may not always reach the expected position.

Episodes dictate how current decisions impact the future

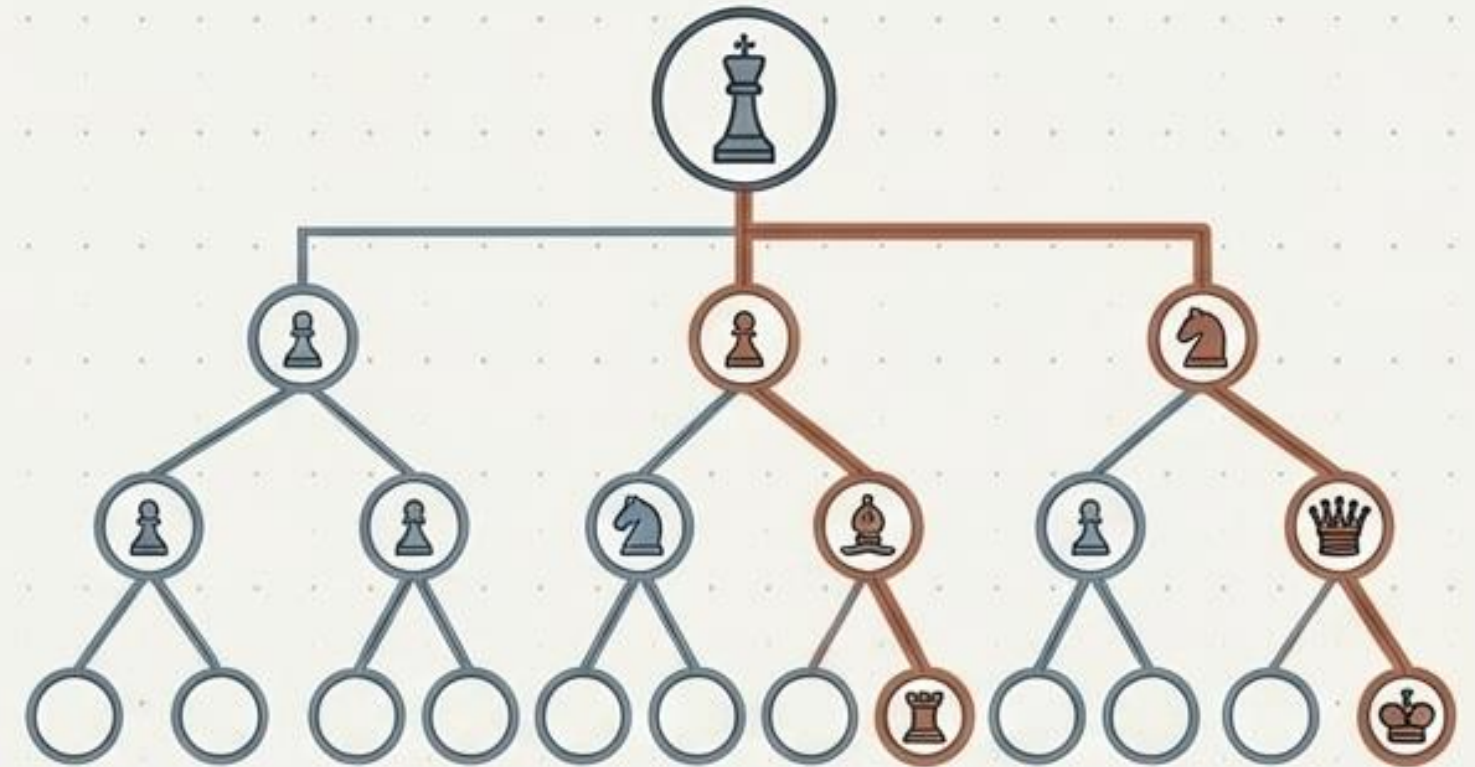
Episodic Environment



Each decision is independent of previous decisions.

Example: Image Classification
(Each image is processed independently).

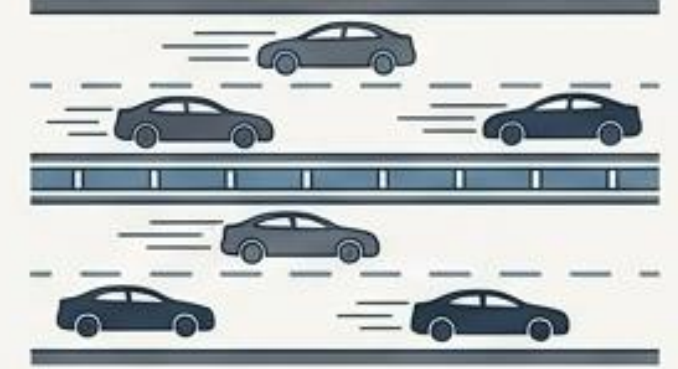
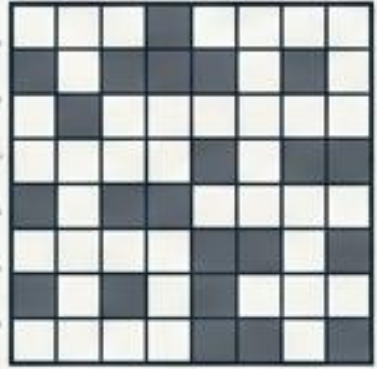
Sequential Environment



Current decisions affect future decisions.

Example: Chess
(Every move influences future moves).

The dynamism spectrum maps environmental stability against time



Static Environment

The environment does not change while the agent is thinking.

Example: Crossword Puzzle
(The puzzle remains the same while you think).

Semi-Dynamic Environment

The environment itself remains unchanged, but the performance measure changes with time. The agent must still act quickly because delays reduce its performance.

Example: Chess with a countdown timer
(The board position remains the same while you think, but your remaining time decreases).

Dynamic Environment

The environment changes continuously while the agent is deliberating.

Example: Self-driving car
(Traffic keeps changing).

Knowledge Check 2: Mastering Semi-Dynamic Environments

Q2: Which of the following is true for semi-dynamic environment?

1. Even if the environment changes with the passage of time while deliberating, the performance score does not change.

2. The environment itself does not change with the passage of time but the agent's performance score does.

3. The environment may change while the agent is deliberating.

4. Environment and performance score, both change simultaneously.

System Diagnostics (Answer Breakdown)

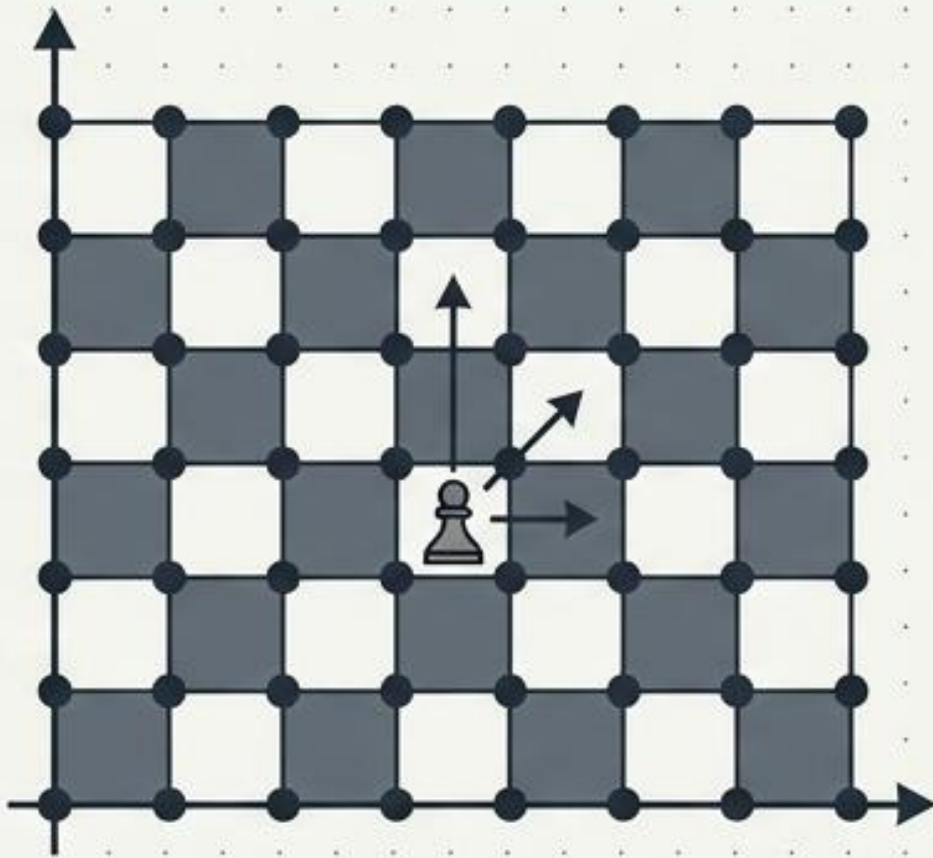
Why 1 is Static: Environment does not change while deliberating. Performance measure remains unchanged. (e.g., Crossword puzzle).

Why 2 is Semi-Dynamic: Environment remains unchanged, but performance measure changes with time. Time decreases continuously, affecting the score. (e.g., Chess with a countdown timer). **[CORRECT]**

Why 3 & 4 are Dynamic: Environment changes continuously; agent must react in real time. (e.g., Self-driving car in moving traffic).

State Space defines how the agent measures the world

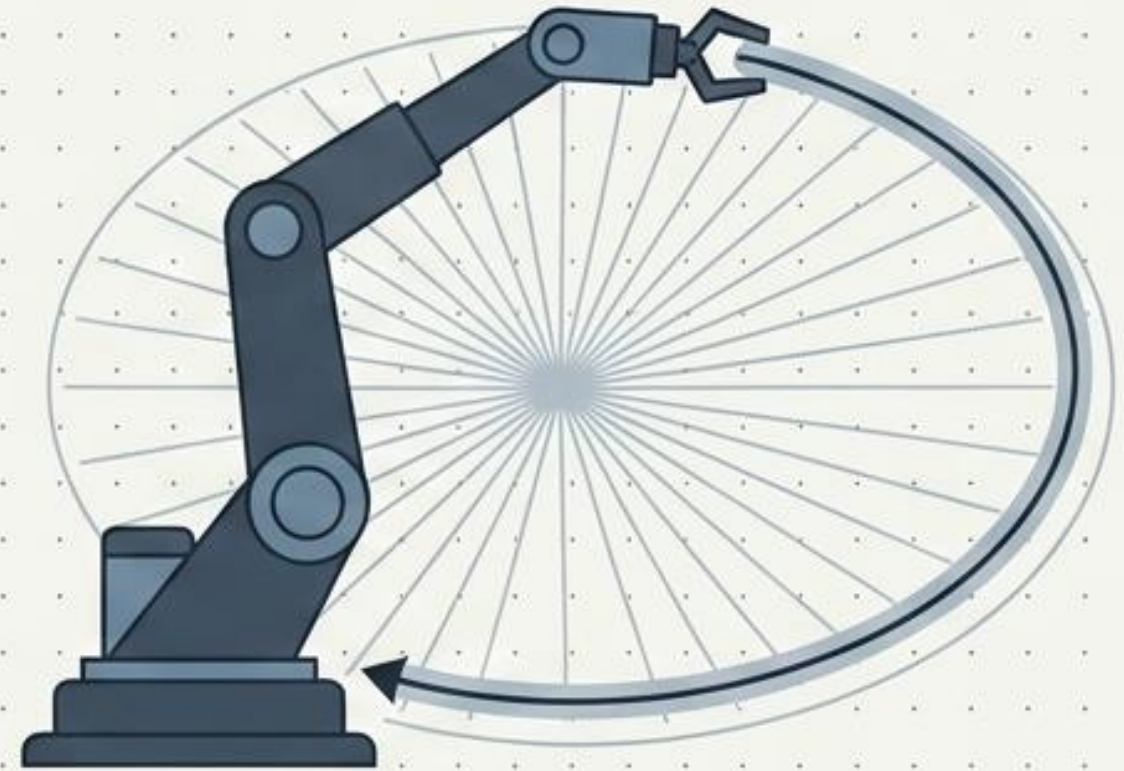
Discrete Environment



Finite number of states and actions.

Example: Chess

Continuous Environment



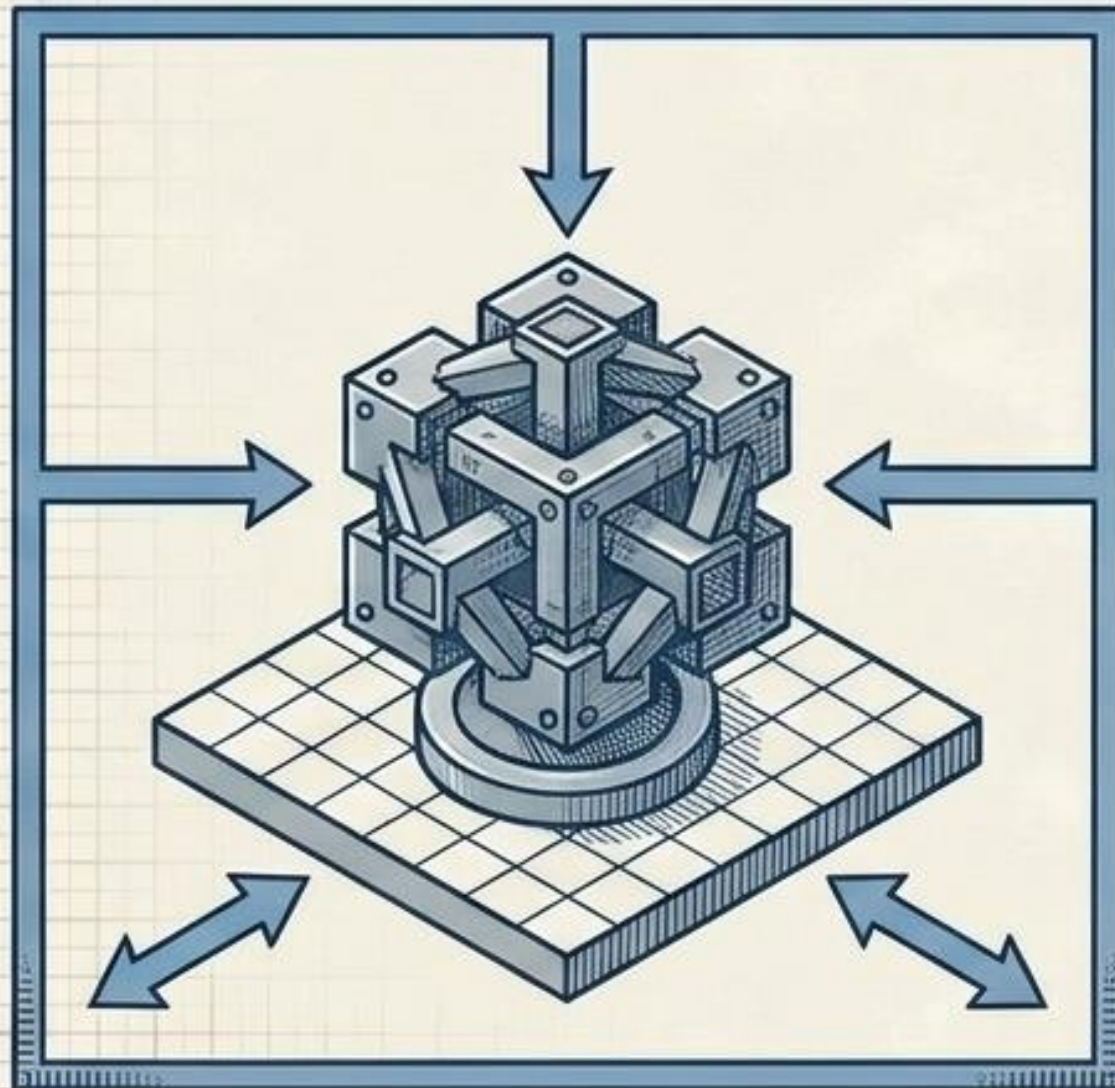
Infinite possible states or values.

Example: Robot movement

Single Agent

Only one intelligent decision-maker exists in the environment.

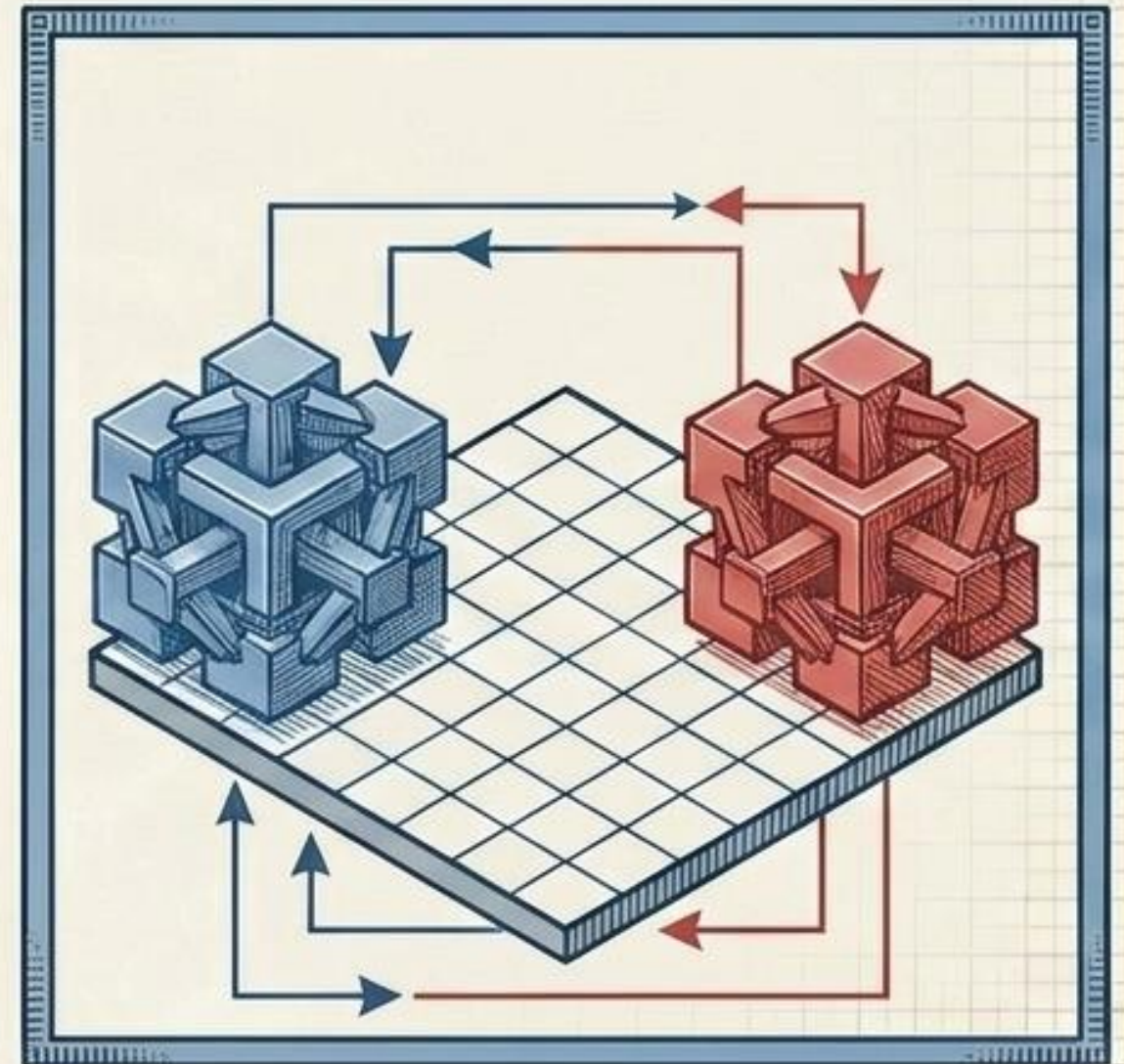
Example:
Sudoku Solver



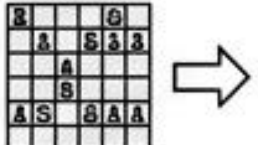
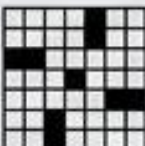
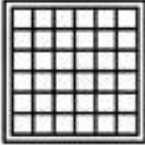
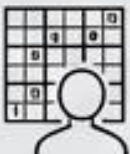
Multi-Agent

Two or more intelligent agents interact, requiring multiple.

Example:
Chess (two players acting as intelligent agents)



The AI Environment Master Matrix

Observability	Fully Observable (Complete info, Chess) 	Partially Observable (Incomplete info, Fog driving) 
Determinism	Deterministic (Predictable, $4+4=8$) 	Stochastic (Uncertain/Random, Stock/Dice) 
Episodes	Episodic (Independent, Image sorting) 	Sequential (Current affects future, Chess) 
Change	Static (No change, Crossword) 	Semi-Dynamic (Performance changes, Timer Chess)  / Dynamic (Continuous change, Self-driving car) 
State Space	Discrete (Finite states, Chess) 	Continuous (Infinite states, Robot arm) 
Agents	Single Agent (One decision maker, Sudoku) 	Multi-Agent (Interacting decision makers, Chess) 